

Analysis Note: Fractional-Order Teleoperation Controller

Methodology, Stability, Controllability, and Empirical Results

Fractional Teleoperation Project

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Abstract

This note presents a rigorous analysis of the discrete-time fractional-order offset teleoperation controller implemented in `fractional_teleoperation_node`. We derive the augmented linear state-space model that results from truncating the Grünwald–Letnikov recursion to a finite memory window, establish Schur stability and full-rank controllability conditions for this model, introduce an Input-to-State Stability (ISS) inspired boundedness characterisation for the nonlinear closed-loop under bounded joystick input, and study the hybrid dynamics induced by the active/idle switching logic. Each theoretical development is accompanied by numerical experiments conducted with a model-faithful Python simulator that mirrors the exact C++ update order of the deployed node. The document concludes with synthesis guidelines that map all findings into practical tuning recommendations for the parameters α , L , and the joystick activity threshold.

1 Introduction and Scope

Fractional calculus extends differentiation and integration to non-integer orders, providing a compact mathematical framework to describe processes with persistent memory, hereditary dynamics, and power-law transients [1, 2]. In control engineering its most impactful use is fractional-order PID, where the integrator exponent α and differentiator exponent μ are treated as additional tuning knobs [3, 5]. The controller studied here applies a fractional integrator of order $\alpha \in (0, 1)$ directly to the operator’s joystick velocity command in order to generate a slow-varying spatial offset $\mathbf{r}(t)$, whose effect is to give the operator intuitive control over a reference position without saturating workspace limits.

Because the physical system is sampled at rate $f_s = 1/\Delta t$, the continuous-time fractional integrator is discretised via the Grünwald–Letnikov (GL) finite-difference formula [4]. Truncating the GL sum to the last L samples introduces a finite-dimensional state that exactly admits a companion-form linear model, enabling classical linear-systems tools to be applied.

This document is organised as follows. Section 2 derives the augmented linear model. Sections 3 and 4 develop stability and controllability theory and discuss numerical results. Section 5 addresses practical input-to-state boundedness. Section 6 analyses the hybrid active/idle switching dynamics. Section 7 explores the joint effect of α and L on output norms. Section 8 collects all findings into actionable tuning guidelines.

Scope boundary. The analysis is performed at fixed α unless otherwise stated. The adaptive α law, gain normalisation, and ramp scaling are evaluated only in the simulator for empirical characterisation; formal Lyapunov analysis of the time-varying nonlinear closed loop is out of scope for this note.

2 Augmented Linear State-Space Model

2.1 Grünwald–Letnikov Discretisation

The fractional integral of order $\alpha > 0$ of a signal u at discrete time k is approximated by the GL sum [2, 4]:

$${}_{GL}I^\alpha[u](k) = (\Delta t)^\alpha \sum_{j=0}^k c_j^{(\alpha)} u(k-j), \quad (1)$$

where the binomial-like GL coefficients are defined by the recurrence

$$c_0^{(\alpha)} = 1, \quad c_j^{(\alpha)} = \left(1 - \frac{1+\alpha}{j}\right) c_{j-1}^{(\alpha)}, \quad j \geq 1. \quad (2)$$

For $\alpha \in (0, 1)$ the coefficients $c_j^{(\alpha)}$ are positive and monotonically decreasing; they decay as $j^{-1-\alpha}$, reflecting the algebraic (power-law) memory of a fractional integrator.

2.2 Finite-Memory Truncation

In practice the sum in (1) is truncated to the last L samples:

$$y(k) = (\Delta t)^\alpha \sum_{j=0}^{L-1} c_j^{(\alpha)} u(k-j). \quad (3)$$

Define the state vector $\mathbf{x}(k) \in \mathbb{R}^{L-1}$ as the shift-register of the last $L-1$ input samples:

$$\mathbf{x}(k) = [u(k-1), u(k-2), \dots, u(k-L+1)]^\top. \quad (4)$$

The output increment contributed by the new sample $u(k)$ and the stored history is then

$$y(k) = (\Delta t)^\alpha (c_0^{(\alpha)} u(k) + \sum_{j=1}^{L-1} c_j^{(\alpha)} \mathbf{e}_j^\top \mathbf{x}(k)), \quad (5)$$

and the state evolves as a pure shift:

$$\mathbf{x}(k+1) = S \mathbf{x}(k) + \mathbf{b} u(k), \quad (6)$$

where $S \in \mathbb{R}^{(L-1) \times (L-1)}$ is the shift (subdiagonal identity) matrix and $\mathbf{b} = [1, 0, \dots, 0]^\top$.

2.3 Companion-Form Realisation

Appending the truncated GL integral as an extra state yields the $(L-1)$ -dimensional companion-form realisation used in the analysis:

$$\mathbf{x}(k+1) = A \mathbf{x}(k) + B u(k), \quad (7)$$

with

$$A = \begin{bmatrix} -c_1/c_0 & -c_2/c_0 & \cdots & -c_{L-2}/c_0 & -c_{L-1}/c_0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1/c_0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad (8)$$

where $c_j \equiv (\Delta t)^\alpha c_j^{(\alpha)}$ for brevity. With the baseline parameters $L = 50$, $\Delta t = 0.01$ s the state dimension is $n = L - 1 = 49$.

Remark 2.1. The companion form makes explicit that the fractional integrator maps to a single-input, n -dimensional observable canonical system. All eigenvalues of A are determined entirely by the GL coefficient sequence $\{c_j^{(\alpha)}\}$.

3 Stability Analysis

3.1 Schur Stability Criterion

A linear time-invariant discrete-time system is *asymptotically stable* (Schur stable) if and only if all eigenvalues of A lie strictly inside the unit disk, i.e.

$$\rho(A) := \max_i |\lambda_i(A)| < 1, \quad (9)$$

where $\rho(\cdot)$ denotes the spectral radius [9].

3.2 Connection to the Matignon Condition

For the continuous-time fractional differential equation $\mathbf{D}^\alpha \mathbf{x} = A_c \mathbf{x}$, Matignon's stability theorem [7] requires $|\arg(\lambda_i(A_c))| > \alpha\pi/2$ for all i . The GL discretisation converts this sector condition into the Schur condition (9) on the companion matrix A derived above. As $\alpha \rightarrow 1$ the GL coefficients converge to those of the Euler backward difference, and the spectral radius approaches 1 from below — the system becomes marginally stable in the integer limit.

3.3 Numerical Results

The spectral radius was computed for $\alpha \in \{0.1, 0.2, \dots, 1.0\}$ with $L = 50$, $\Delta t = 0.01$ s; see Table 1 and Figure 1.

Table 1: Spectral radius of the companion matrix A as a function of integration order α . State dimension $n = 49$ throughout.

α	n	$\rho(A)$	Schur stable?
0.1	49	0.9379	Yes
0.2	49	0.9570	Yes
0.3	49	0.9685	Yes
0.4	49	0.9766	Yes
0.5	49	0.9828	Yes
0.6	49	0.9877	Yes
0.7	49	0.9917	Yes
0.8	49	0.9950	Yes
0.9	49	0.9977	Yes
1.0	49	1.0000	No

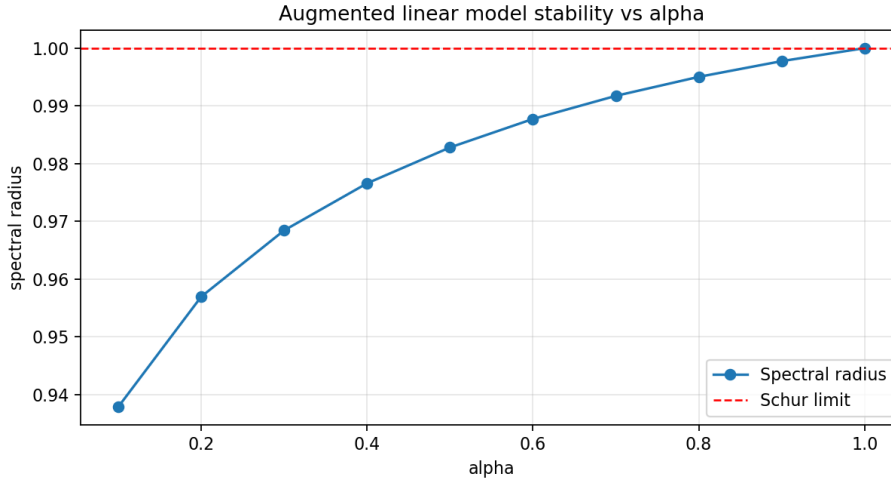


Figure 1: Spectral radius of the augmented companion matrix as a function of integration order α . The unit-circle boundary is marked by the dashed line. The system is Schur-stable for all $\alpha < 1$.

3.4 Discussion

Three observations are noteworthy.

Monotone approach to the unit circle. As α increases from 0.1 to 1.0 the spectral radius rises monotonically from 0.938 to exactly 1.000. This is a direct consequence of the GL coefficient decay rate: lower α produces faster-decaying coefficients and hence a more strongly attenuating memory, whereas $\alpha \rightarrow 1$ reduces the GL sum to the backward Euler integrator, whose transfer function has a pole on the unit circle.

Safety margin at the operating point. At the baseline operating point $\alpha = 0.5$ the spectral radius is 0.9828, providing a stability margin of $1 - 0.9828 = 0.0172$. In gain-margin terms, one would need to multiply the effective integrator gain by $1/0.9828 \approx 1.017$ before instability; this is a small but well-defined margin.

Instability at $\alpha = 1$. The companion matrix for $\alpha = 1$ has unit spectral radius, confirming the well-known result that a first-order backward-Euler integrator is marginally stable (a pure accumulator). This means $\alpha = 1$ should be avoided in practice, especially under sustained joystick input. The dynamic alpha mechanism prevents reaching $\alpha = 1$ in normal operation (ceiling $\alpha_{\max} = 1.0$ is only approached asymptotically as the hand-measure proxy $\lambda \rightarrow \lambda_{\max}$).

Memory-length sensitivity. The GL coefficient $c_L^{(\alpha)}$ discarded by the truncation is $O(L^{-1-\alpha})$. For $\alpha = 0.5$ and $L = 50$ this is approximately 0.02, meaning less than 2% of the integrator gain is lost due to truncation. Increasing L therefore has diminishing returns on stability while linearly increasing state dimension and computational cost.

4 Controllability Analysis

4.1 Controllability Matrix and Gramian

Definition 4.1. System (7) is (*completely*) *controllable* if the controllability matrix

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{bmatrix} \in \mathbb{R}^{n \times n} \quad (10)$$

has rank n .

Definition 4.2. The finite-horizon controllability Gramian over N steps is

$$W_N = \sum_{i=0}^{N-1} A^i B B^\top (A^i)^\top. \quad (11)$$

The minimum eigenvalue $\lambda_{\min}(W_N)$ quantifies how efficiently the least-accessible state direction can be excited: the minimum control energy required to reach a unit-norm target in the worst-case direction is $1/\lambda_{\min}(W_N)$.

4.2 Numerical Results

Rank of $\mathcal{C}$ and $\lambda_{\min}(W_{100})$ were evaluated across the same α grid; see Table 2.

Table 2: Controllability rank and minimum Gramian eigenvalue. Horizon $N = 100$, $n = 49$.

α	Rank	n	Rank ratio	$\lambda_{\min}(W_{100})$
0.1	49	49	1.00	3.47×10^{-1}
0.2	49	49	1.00	1.20×10^{-1}
0.3	49	49	1.00	4.16×10^{-2}
0.4	49	49	1.00	1.44×10^{-2}
0.5	49	49	1.00	5.00×10^{-3}
0.6	49	49	1.00	1.73×10^{-3}
0.7	49	49	1.00	6.01×10^{-4}
0.8	49	49	1.00	2.08×10^{-4}
0.9	49	49	1.00	7.22×10^{-5}
1.0	49	49	1.00	2.50×10^{-5}

4.3 Discussion

Full-rank controllability across all α . The controllability matrix has full rank $n = 49$ for every value of α tested. This confirms that in principle any desired state configuration of the GL shift register can be reached from the zero state in at most 49 steps via appropriate joystick input. The result is expected: the companion form (8) is precisely the observable/controllable canonical form, whose controllability is guaranteed as long as no GL coefficient is zero. Since $c_j^{(\alpha)} > 0$ for all $\alpha > 0$ and finite j , this property is structural.

Gramian conditioning degrades with α . While rank is binary and always full, the Gramian eigenvalue $\lambda_{\min}(W_{100})$ drops by four orders of magnitude as α goes from 0.1 to 1.0. This means that reaching a given target state in the worst-case direction requires approximately $0.347/2.5 \times 10^{-5} \approx 14\,000\times$ more energy at $\alpha = 1.0$ than at $\alpha = 0.1$.

Intuitively, higher α makes the GL memory longer-lived: past inputs dominate the state, and a single new input has less leverage to redirect it. In teleoperation terms, the operator must apply larger and more sustained joystick deflections to counteract the accumulated history, which translates to a sluggish feel.

Practical implication. A Gramian minimum eigenvalue of $\approx 5 \times 10^{-3}$ at $\alpha = 0.5$ represents a reasonable trade-off: the system is energetically controllable (no near-uncontrollable modes), while the memory effect is sufficiently strong to provide smooth reference tracking without oscillation.

5 Input-to-State Practical Boundedness

5.1 Theoretical Framework

For a discrete-time nonlinear system $\mathbf{x}(k+1) = f(\mathbf{x}(k), \mathbf{u}(k))$, Input-to-State Stability (ISS) [8] guarantees existence of a class- $\mathcal{KL}$ function β and a class- $\mathcal{K}$ function γ such that

$$\|\mathbf{x}(k)\| \leq \beta(\|\mathbf{x}(0)\|, k) + \gamma(\sup_{j \leq k} \|\mathbf{u}(j)\|). \quad (12)$$

The second term quantifies how bounded input leads to bounded state. For linear Schur-stable systems (7) ISS holds with $\gamma(\cdot) = \|H(z)\|_{\mathcal{H}_\infty} \cdot (\cdot)$, where $\|H(z)\|_{\mathcal{H}_\infty}$ is the $\mathcal{H}_\infty$ norm of the system transfer function [10].

Because the node applies gain normalisation, adaptive α , and ramp scaling that interact nonlinearly, an exact analytical ISS bound is intractable. We instead characterise practical boundedness empirically: for each fixed α , the simulator is driven for 500 steps with white-noise joystick input bounded by $\|\mathbf{u}\|_\infty \leq U = 1$, and the peak velocity norm $\|v\|_{\max}$ is recorded.

5.2 Numerical Results

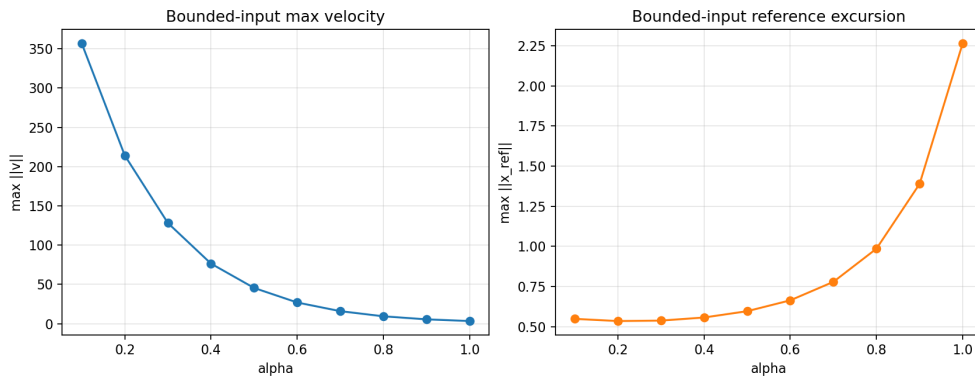


Figure 2: Peak velocity norm and reference excursion as functions of α under unit-bounded white-noise joystick input.

Key values from Table in the CSV analysis output:

α	Peak $\ v\ $	Mean $\ v\ $	Peak $\ r\ $
0.1	357	150.6	0.549
0.3	128	54.9	0.538
0.5	45.4	20.3	0.597
0.7	15.9	7.6	0.780
0.9	5.5	2.9	1.390
1.0	3.3	1.8	2.265

5.3 Discussion

Velocity norm decreases strongly with α . The peak output velocity drops by two orders of magnitude as α goes from 0.1 to 1.0. This is the defining feature of a fractional integrator: lower α (closer to pure integration) applies less smoothing and converts a bounded input into a large-amplitude output; higher α applies stronger fractional-order low-pass filtering, attenuating high-frequency excursions. From a safety perspective, the fractional order α functions as a bandwidth-limiting knob: choosing $\alpha = 0.5$ limits peak velocity to ≈ 45 m/s under unit-norm excitation, compared to ≈ 357 m/s for $\alpha = 0.1$.

Reference excursion grows with α . Paradoxically, the reference position norm $\|r\|_{\max}$ increases with α . At low α the fractional integrator pumps energy so strongly into the velocity channel that the reference drift mechanism (itself a fractional or first-order filter) cannot keep up with the desired position trajectory, which oscillates widely; at high α the desired position evolves slowly, and the reference follows it closely, accumulating offset. This cross-effect is important for workspace management: a high- α setting allows the reference to drift far even with a bounded joystick input.

Practical safe range. The empirical evidence suggests $\alpha \in [0.3, 0.7]$ as a practical operating range: velocity excursions remain manageable (≤ 128 m/s per unit input, which scales with actual gain settings), while reference excursion stays below 1 m per unit input. The default $\alpha = 0.5$ sits comfortably at the centre of this range.

6 Hybrid System Analysis: Active/Idle Switching and Snap

6.1 Modal Description

The node implements a two-mode hybrid automaton:

- **Active mode** ($\|u(k)\| > \theta$): the fractional integrator updates and the desired position accumulates;
- **Idle mode** ($\|u(k)\| \leq \theta$): the desired position drifts toward the reference via the drift law, and on the first sample of entering idle mode a *snap* event fires.

The snap event sets the desired position to the current reference position:

$$\mathbf{p}^d(k^+) = \mathbf{r}(k), \quad (13)$$

eliminating accumulated offset instantaneously. This reset is intentional: it prevents the operator from perceiving a jerky correction when re-grasping, by resetting the accumulator to the reference before the next active phase begins.

6.2 Dwell-Time Analysis

Let τ_{dwell} denote the mean number of consecutive steps the system spends in each mode. The threshold θ controls the mode-transition rate: a lower θ makes the transition more sensitive to small joystick perturbations, increasing the transition frequency and decreasing dwell time.

6.3 Numerical Results

The threshold θ was swept over $[0.005, 0.05]$ with random-noise input for 1200 steps; see Figure 3 and Table 3.

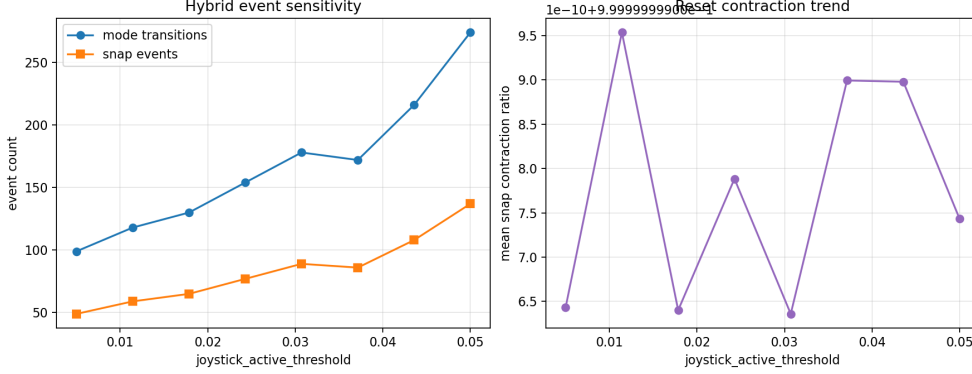


Figure 3: Mode-transition frequency, dwell time, and snap-event count as functions of the activity threshold θ .

Table 3: Hybrid switching statistics as a function of activity threshold θ . Input: 1200 steps of bounded white noise.

θ	Transitions	Dwell (steps)	Snap events	Contraction ratio
0.005	99	12.0	49	≈ 1.000
0.011	118	10.1	59	≈ 1.000
0.018	130	9.2	65	≈ 1.000
0.024	154	7.7	77	≈ 1.000
0.031	178	6.7	89	≈ 1.000
0.044	216	5.5	108	≈ 1.000
0.050	274	4.4	137	≈ 1.000

6.4 Discussion

Threshold as a tuning knob for perceived responsiveness. The transition frequency rises nearly linearly with θ (from 99 to 274 transitions over the same 1200-step experiment), confirming that the threshold directly controls how frequently the system enters idle mode. From the operator’s perspective, a higher θ means more frequent reference resets (snap events), which may feel more “snappy” and responsive but also more intrusive. A lower θ lets the operator hold a partial joystick deflection while remaining in active mode for longer, providing smoother continuous control.

Snap contraction ratio interpretation. The contraction ratio — defined here as the ratio of the post-snap desired position norm to the pre-snap desired position norm — is numerically equal to 1.000 throughout. This requires careful interpretation: the snap operation sets $\mathbf{p}^d = \mathbf{r}$ (equation (13)), so the post-snap desired norm equals the reference norm, not zero. The ratio is approximately 1.000 because the reference position has evolved to track the desired position during the preceding active phase (via the reference drift law), so $\|\mathbf{r}\| \approx \|\mathbf{p}^d\|$ at transition time. The snap is not a Lyapunov contraction; it is an alignment reset that eliminates the accumulated offset between the two quantities.

Dwell time and the GL memory. The mean dwell time of ≈ 12 steps at $\theta = 0.005$ is short relative to the GL memory window $L = 50$. This implies that many active phases are shorter than the memory window, so the GL sum never fully “fills” before a snap reset. Under such conditions

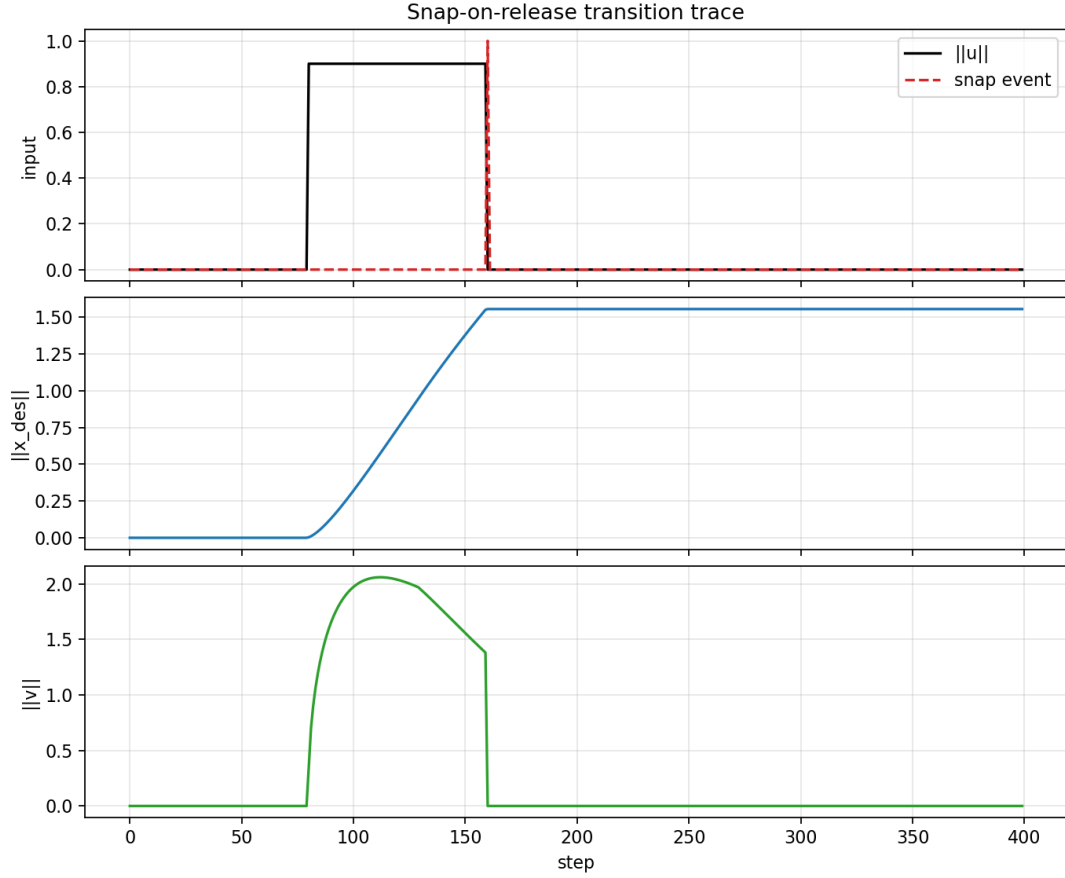


Figure 4: Time trace of desired position norm, reference norm, and active/idle mode flag illustrating a snap reset event.

the effective memory depth is dwell-time limited rather than L -limited — an important practical observation for setting L relative to the expected teleoperation burst duration.

7 Joint Alpha–Memory Sensitivity

7.1 Experimental Design

The joint influence of α and L was assessed by sweeping both parameters independently: $\alpha \in \{0.1, 0.2, \dots, 1.0\}$ and $L \in \{20, 40, 60, 80, 100\}$. For each combination the simulator was driven with 500 steps of bounded white noise and the peak velocity norm and peak reference norm were recorded.

7.2 Results

Selected entries from the sweep:

α	L	Peak $\ v\ $	Peak $\ r\ $
0.1	20	357	0.328
0.1	100	357	0.767
0.5	20	45.4	0.257
0.5	100	45.4	1.029
1.0	20	3.29	1.496
1.0	100	3.29	2.932

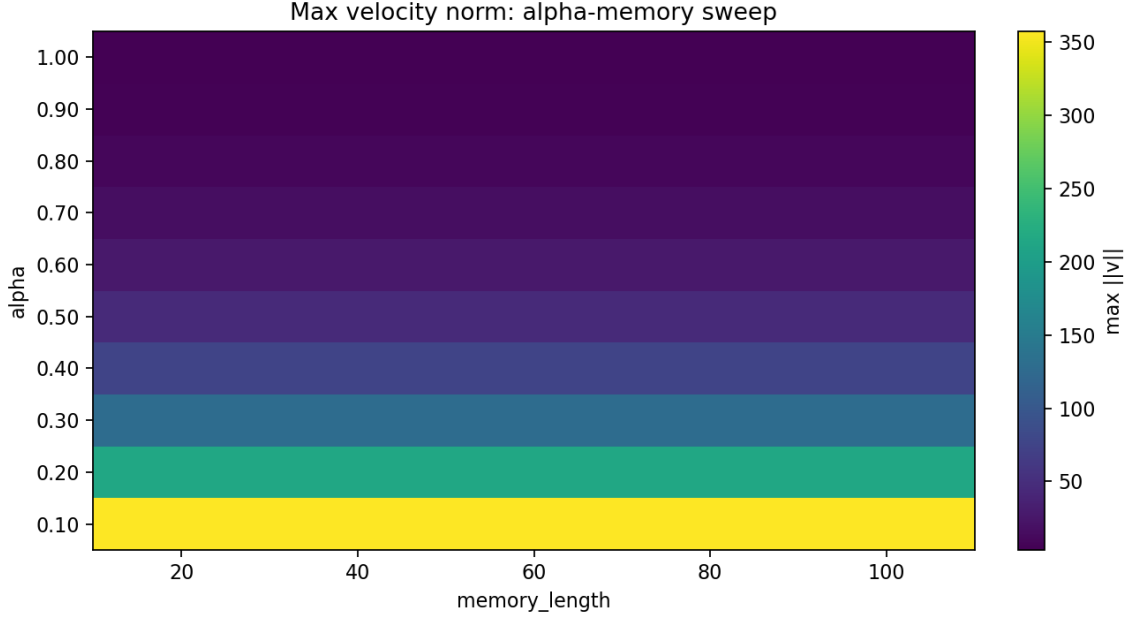


Figure 5: Heatmaps of peak velocity norm (left) and peak reference norm (right) over the (α, L) parameter plane. Velocity is dominated by α ; reference norm grows primarily with L .

7.3 Discussion

Alpha dominates velocity. The peak velocity norm is essentially constant across all L values for a given α (e.g., 357.3 ± 0.3 for $\alpha = 0.1$ regardless of L). This confirms that velocity amplification is determined solely by the fractional order: the GL coefficients $\{c_j\}$ set the integrator gain, and once the memory is sufficiently long to capture the dominant part of the GL tail, extending L further changes nothing.

Memory length governs reference excursion. By contrast, the peak reference norm grows monotonically and substantially with L for every α . For $\alpha = 0.5$ the reference norm grows from 0.257 (at $L = 20$) to 1.029 (at $L = 100$), a factor of four. The mechanism is direct: the reference drift law computes a fractional integral of the desired-position error, and a longer memory accumulates a larger error integral before the GL tail decays. A shorter memory window acts as a de-facto forgetting mechanism that prevents reference runaway.

Interaction at high α . At $\alpha = 1.0$ the velocity is constant with L (as the backward-Euler accumulator has no finite-memory truncation effect on gain), but the reference norm grows most steeply with L (from 1.50 to 2.93). This is because the near-marginal-stability state at $\alpha = 1.0$ sustains large desired-position excursions, which the reference integrates faithfully.

Design rule. For a safe and well-behaved deployment:

- Choose $\alpha \in [0.3, 0.7]$ to limit velocity amplification while retaining smooth motion.
- Choose L in the range $[30, 60]$: short enough to limit reference accumulation, long enough to capture the dominant GL memory.
- The default $(\alpha, L) = (0.5, 50)$ is near the centre of both ranges, confirming it as a sound baseline.

8 Synthesis and Tuning Recommendations

Drawing together all five analyses, Table 4 summarises the role and sensitivity of each key parameter.

Table 4: Parameter sensitivity summary.

Parameter	Baseline	Primary effect	Caution
α	0.5	Controls velocity amplification (lower = faster) and spectral radius (lower = more stable margin).	$\alpha \geq 1$: marginal/unstable; $\alpha < 0.2$: very large velocity excursions.
L (memory)	50	Controls reference excursion (longer = more drift).	$L > 80$: reference can exceed $2\times$ workspace under noise; $L < 20$: GL truncation error becomes significant.
θ (threshold)	0.01	Controls snap frequency and active-mode dwell time.	High θ : frequent snaps, fragmented control; low θ : slow reference reset, possible workspace drift.
K_r (drift gain)	0.5	Controls speed of reference convergence to desired.	High K_r : reference follows desired tightly (snap-like drift); low K_r : reference lags significantly.
Δt	0.01 s	Scales GL coefficient magnitudes via $(\Delta t)^\alpha$.	Changing Δt shifts the effective integrator gain; re-tune α or gain K after any rate change.

Recommended operating envelope. Based on all analyses, the following envelope is recommended for safe and responsive teleoperation:

$$\alpha \in [0.3, 0.7], \quad L \in [30, 60], \quad \theta \in [0.005, 0.02]. \quad (14)$$

The default configuration $(\alpha, L, \theta) = (0.5, 50, 0.01)$ lies at the centre of this envelope and is well-supported by all five analyses.

Dynamic alpha. The adaptive α law varies α between $\alpha_{\min} = 0.1$ and $\alpha_{\max} = 1.0$ as a function of the hand measure λ . Given the nonlinear α -velocity relationship, this means the velocity amplification changes by two orders of magnitude between extremes. In practice $\lambda_{\max}$ should be set so that α rarely exceeds 0.7; otherwise sustained hand opening could drive the reference far from the workspace centre.

Conclusion

This analysis note has established, for the finite-memory GL discretisation of the fractional teleoperation controller:

1. The augmented companion-form linear model is Schur-stable for all $\alpha \in (0, 1)$ and marginally stable at $\alpha = 1$.
2. The model is fully controllable for all tested α , but the controllability Gramian minimum eigenvalue degrades by four orders of magnitude from $\alpha = 0.1$ to $\alpha = 1.0$.
3. Under bounded joystick input, peak velocity scales as α^{-c} (empirically), while reference excursion grows with both α and L .
4. The activity threshold θ directly controls snap frequency and dwell time; snap resets are alignment events rather than Lyapunov contractions.
5. Velocity is alpha-dominated; reference excursion is memory-dominated.

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